

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA0409

COURSE NAME: Operating Systems

COURSE OUTCOME COVERED

CO4: Optimize various file allocation strategies and disk scheduling algorithms to identify optimal methods for enhancing system performance.

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

INDUSTRY PROBLEM SOLVING TASK

Code of Conduct:

I k.jaya venkata sai / 192525182 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Subject : OS

Code : CSA0409

C04- ATQ

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1. File Allocation - Contiguous, Linked & Indexed
Allocation

For a cloud storage Company Experiencing frequent file access delays, Contiguous allocation would be the most suitable and access is important.

* Contiguous allocation : Stores a file in consecutive disk blocks. It provides very fast access because the blocks are physically close together.

* Linked allocation : Stores file blocks anywhere on the disk and links them using pointer pointers. It avoids external fragmentation but is slower for direct access because pointers must be followed.

7 Indexed allocation : Uses an Index block containing pointers to all file blocks.

Best choice : - Contiguous Allocation.

Conclusion : - for high performance cloud storage with mostly stable file sizes, Contiguous allocation gives better access performance.

2.

Disk Scheduling - FCFS, SSTF and SCAN.

The data center experiences high disk head movement while processing user requests.

- FCFS : Services requests in their arrival order, it is simple but can cause large head movements.
- SSTF : Services the requests closest to the current head position. It reduces immediate seek time but may cause starvation of distant requests.
- SCAN : moves the disk head in one direction servicing requests along the way, and then reverses direction.

Why scan is suitable :

1. Reduces unnecessary disk-head movement.
2. Provides better average waiting time than FCFS.
3. Avoids starvation better than SSTF.
4. Gives more predictable performance.

Conclusion : SCAN is suitable for a busy data center because it balances performance and fairness.

3.

Bitmap - Based free Space Management.

A Bit map represents each disk block using a bit :

- * 0 $\rightarrow$ Block is free
- * 1 $\rightarrow$ Block is allocated.

When the disk size increases, a very large bitmap must be searched, which can make allocation slower.

Proposed optimization : Hierarchical Bitmap.

A hierarchical bitmap can be used instead of searching the complete bitmap every time.

Working :-

1. Divide the disk blocks into groups.
2. Maintain a bitmap for each group.
3. Maintain another higher-level bitmap indicating which groups contain free blocks.
4. First search the higher-level bitmap.
5. Search only the selected group for a free block.

Advantages :-

- Reduces Search Time.
- Improves allocation speed for large disks.
- Requires fewer bitmap entries to be scanned during each allocation.

Conclusion :- Hierarchical bitmap-based allocation improves the efficiency and scalability.

4.

Scan Vs C-Scan for Real Time Applications

Feature	Scan	C-Scan.
Head Movement	Both directions	one servicing direction
waiting time	Less uniform	more uniform
Predictability	Good	Better
Fairness	Good	Very Good
Real-time suitability	moderate	High

C-Scan Working :- The disk Head moves in one direction and services all requests on the way. When it reaches the end, it returns directly to the beginning without servicing requests during the return trip.

Why C-Scan is Preferred :

- Provides more uniform waiting time.
- Gives predictable service to requests.
- Reduces the Variation in response time.
- Suitable for systems where consistent performance is important.

Hybrid storage system - file allocation + Disk Scheduling

A large scale storage system can combine Indexed Allocation with SCAN disk scheduling.

Proposed Combination :

File Allocation : Indexed Allocation.

- Each file has an index block.
- The index block contains addresses of the file's data blocks.
- files can grow dynamically.

Disk scheduling : SCAN.

- The disk head moves in one direction.
- Requests encountered along the path.
- The head reverses direction after reaching the end.

Conclusion : An Indexed allocation + SCAN hybrid system is a practical choice for large-scale storage because indexed allocation efficiently manages files while SCAN efficiently manages disk access requests.